

- (d) A 0.2% solution of phenol is antiseptic while 1% solution acts as disinfectant.
 (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant. Chlorine and Iodine are used as strong disinfectants. Dilute solution of boric acid and hydrogen peroxide are weak antiseptics. Disinfectants harm the living tissues. Hence, statement (b) is not true.

ORGANIC CHEMISTRY

A. Hydrocarbons

1. Which is the fundamental element of all organic compounds?

- (a) Nitrogen (b) Oxygen
 (c) Carbon (d) Brimstone

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Carbon is the fundamental element of all organic compounds (including bio-compounds). The organic compounds are a large class of chemical compounds in which one or more atoms of carbon are covalently linked to atoms of other elements, most commonly hydrogen, oxygen or nitrogen.

2. A hydrocarbon in which two carbon atoms are joined by a double bond is called as an :

- (a) Alkane (b) Alkene
 (c) Alkyne (d) Ionic bond
 (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

A hydrocarbon in which two carbon atoms are joined by a double bond is called as an alkene. Alkenes have the general formula C_nH_{2n} . They are also known as olefins. Ethylene (C_2H_4), Propylene (C_3H_6) and Butylene (C_4H_8) are first three members of this group.

3. Which of the following gases is used in cigarette lighters?

- (a) Butane (b) Methane
 (c) Propane (d) Radon

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Butane is a flammable hydrocarbon with the molecular formula C_4H_{10} . It is a natural gas perhaps best known for its use as a fuel cigarette lighters. It is also an organic compound known as NGL, a Natural Gas Liquid.

4. Gases used in welding are :

- (a) Oxygen and hydrogen (b) Oxygen and nitrogen
 (c) Oxygen and acetylene (d) Hydrogen and acetylene
 (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

Welding is a process of joining two materials, usually metals, by causing coalescence. Acetylene is the most commonly need welding gas it is also used in oxyacetylene welding and cutting. Oxygen is a gas that is used in welding to increase the rate of the welding process.

5. Numbers of sigma and pi bonds in benzene are:

- (a) 3, 3 (b) 3, 6
 (c) 12, 3 (d) 12, 6
 (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Benzene is an important organic chemical compound with chemical formula C_6H_6 . Its molecule is composed of 6 Carbon atoms and 6 Hydrogen atoms. Its chemical structure can be described as a hexagonal ring with alternating double bonds. In benzene there are 6 sigma bonds between carbon atoms and 6 sigma bonds between carbon and hydrogen atoms. It has also 3 pi bonds between carbon atoms. Therefore 12 sigma and 3 pi bonds are present in benzene molecule.

B. Alcohol

1. Fermentation of sugar leads to –

- (a) Ethyl alcohol (b) Methyl alcohol
 (c) Acetic acid (d) Chlorophyll

47th B.P.S.C. (Pre) 2005

Ans. (a)

Ethyl alcohol is formed by the fermentation of sugar, which is made of glucose and fructose.

2. Final product of Fermentation

- (a) Pyruvic acid (b) Acetaldehyde
 (c) Ethyl alcohol (d) Formic acid

46th B.P.S.C. (Pre) 2004

Ans. (c)